

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset
df = pd.read_csv("Titanic-Dataset.csv")

# -----
# Basic Dataset Information
# -----
print("First 5 Rows")
print(df.head())

print("\nLast 5 Rows")
print(df.tail())

print("\nDataset Information")
print(df.info())

print("\nShape of Dataset")
print(df.shape)

print("\nMissing Values")
print(df.isnull().sum())

# -----
# Summary Statistics
# -----
print("\nSummary Statistics")
print(df.describe())

# -----
# Histograms
# -----
df.hist(figsize=(12,10))
plt.suptitle("Histogram of Numerical Features")
plt.show()

# -----
# Boxplots
# -----
numeric_cols = df.select_dtypes(include=np.number).columns

for col in numeric_cols:
    plt.figure(figsize=(5,3))
    sns.boxplot(x=df[col])
    plt.title(f"Boxplot of {col}")
    plt.show()

# -----
# Correlation Matrix
# -----
plt.figure(figsize=(10,8))

corr = df[numeric_cols].corr()

sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Matrix")
plt.show()

# -----
# Pairplot
# -----
sns.pairplot(

```

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df[['Survived', 'Pclass', 'Age', 'Fare']]
)

plt.show()

# -----
# Survival Analysis
# -----
plt.figure(figsize=(6,4))

sns.countplot(
    x="Survived",
    data=df
)

plt.title("Survival Distribution")
plt.show()

# -----
# Gender vs Survival
# -----
plt.figure(figsize=(6,4))

sns.countplot(
    x="Sex",
    hue="Survived",
    data=df
)

plt.title("Gender vs Survival")
plt.show()

# -----
# Passenger Class vs Survival
# -----
plt.figure(figsize=(6,4))

sns.countplot(
    x="Pclass",
    hue="Survived",
    data=df
)

plt.title("Passenger Class vs Survival")
plt.show()

# -----
# Age Distribution
# -----
plt.figure(figsize=(8,5))

sns.histplot(
    df["Age"],
    bins=30,
    kde=True
)

plt.title("Age Distribution")
plt.show()

# -----
# Fare Distribution
# -----
plt.figure(figsize=(8,5))

sns.histplot(
    df["Fare"],
    bins=30,
    kde=True
)

plt.title("Fare Distribution")
plt.show()
```

```
print("\nEDA Completed Successfully")
```


First 5 Rows

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Last 5 Rows

	PassengerId	Survived	Pclass	Name	\
886	887	0	2	Montvila, Rev. Juozas	
887	888	1	1	Graham, Miss. Margaret Edith	
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	
889	890	1	1	Behr, Mr. Karl Howell	
890	891	0	3	Dooley, Mr. Patrick	

	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	male	27.0	0	0	211536	13.00	NaN	S
887	female	19.0	0	0	112053	30.00	B42	S
888	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	male	26.0	0	0	111369	30.00	C148	C
890	male	32.0	0	0	370376	7.75	NaN	Q

Dataset Information

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	object
9	Fare	891 non-null	float64
10	Cabin	204 non-null	object
11	Embarked	889 non-null	object

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

Shape of Dataset

(891, 12)

Missing Values

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

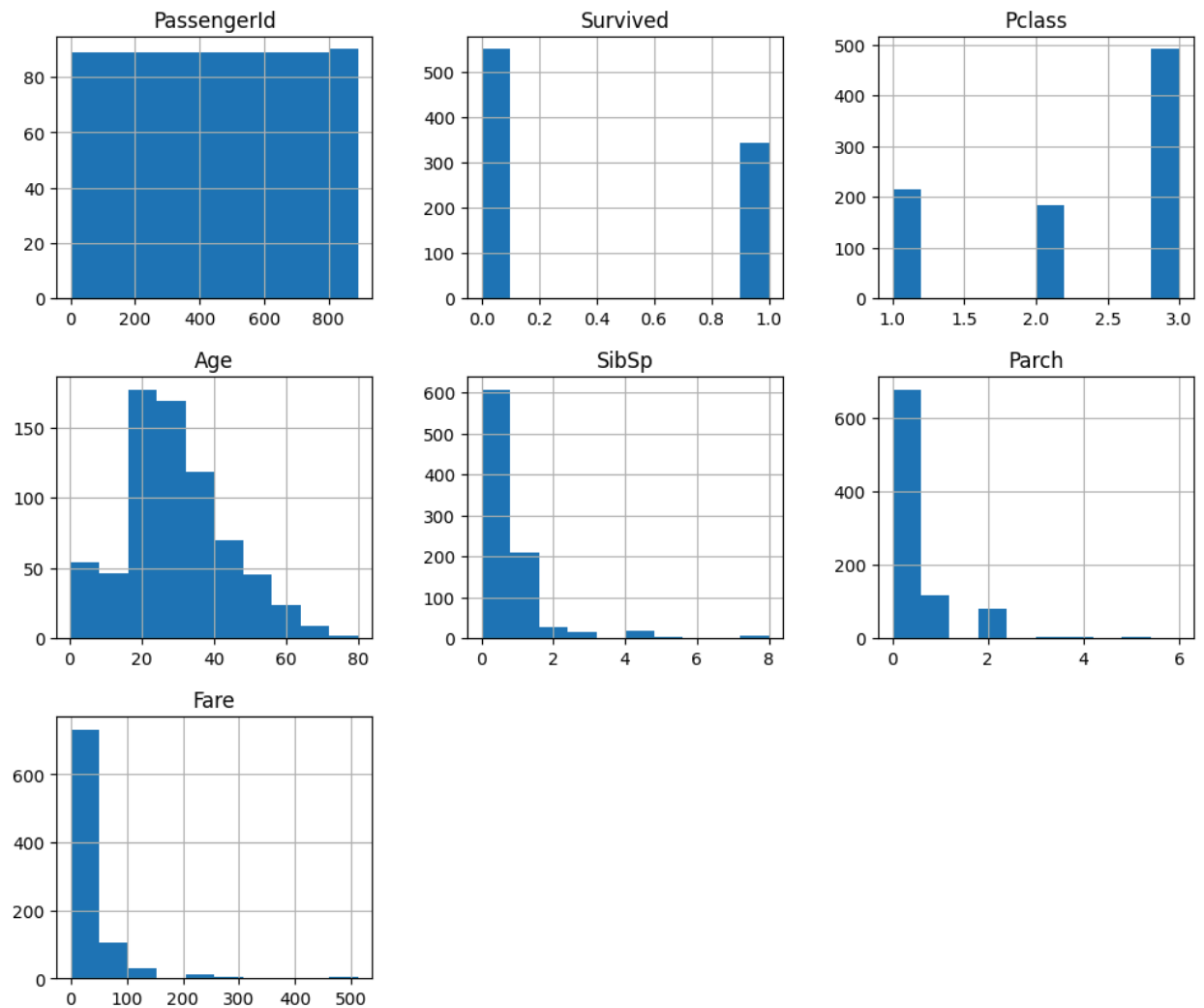
Summary Statistics

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	

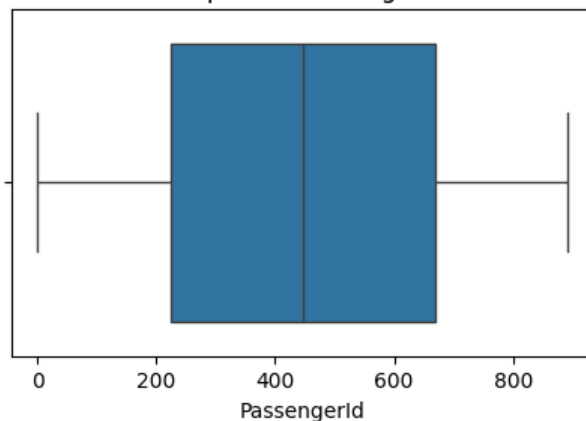
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Histogram of Numerical Features



Boxplot of PassengerId



Boxplot of Survived

